

Python Package testing

Organizing tests inside your package

```
mysklearn/  
|-- mysklearn    <-- package  
|-- tests       <-- tests directory  
|-- setup.py  
|-- LICENSE  
|-- MANIFEST.in
```

Organizing tests inside your package

Test directory layout

```
mysklearn/tests/  
|-- __init__.py  
|-- preprocessing  
|   |-- __init__.py  
|   |-- test_normalize.py  
|   |-- test_standardize.py  
|-- regression  
|   |-- __init__.py  
|   |-- test_regression.py  
|-- test_utils.py
```

Code directory layout

```
mysklearn/mysklearn/  
|-- __init__.py  
|-- preprocessing  
|   |-- __init__.py  
|   |-- normalize.py  
|   |-- standardize.py  
|-- regression  
|   |-- __init__.py  
|   |-- regression.py  
|-- utils.py
```

Running tests with pytest

```
pytest
```

```
===== test session starts =====
platform linux -- Python 3.7.9, pytest-6.1.2, py-1.9.0, pluggy-0.13.1
rootdir: /home/workspace/mypackages/mysklearn
collected 6 items

tests/preprocessing/test_normalize.py ... [ 50%]
tests/preprocessing/test_standardize.py ... [100%]

===== 6 passed in 0.23s =====
```

Testing multiple versions of Python

This `setup.py` allows any version of Python from version 2.7 upwards.

```
from setuptools import setup, find_packages

setup(
    ...
    python_requires='>=2.7',
)
```

To test these Python versions you must:

- Install all these Python versions
- Run `tox`